

Deep Deep Learning

Graduate student-led research seminar | Semester: TBD | Course number: TBD

Instructor: Assaf Shocher. Course contact and office hours: TBD.

Course description

Major ideas in deep learning, studied through foundational material, a central paper and recent developments. The candidate pool includes NTK, Mamba, in-context learning, test-time training, hypernetworks, world models, model merging, interpretability and new generative frameworks.

Topic selection

The 18 topics in this syllabus are candidates, not weekly assignments. Students and the instructor choose which topics enter the final 13-week schedule. Unchosen topics are not assigned. Selection, order and presenter assignments: TBD. Each selected topic gets two academic hours.

Learning objectives

Explain a mathematical mechanism; identify its assumptions; assess an experiment that tests an explanation; connect recent research to older ideas; formulate a precise unresolved question.

Prerequisites & logistics

Working knowledge of deep learning, linear algebra, probability, calculus and optimization. Formal prerequisites, credits, enrollment, meeting time, room and delivery mode: TBD. Dates are not yet assigned. Flow matching is assumed background from the base course.

Seminar format & assessment

Students lead presentations and discussion. Each session centers on one main paper, with selected foundation and companion material. Prepare one derivation, one decisive experiment and one unresolved question. No exam. Grading components and weights, attendance requirements, written submissions, reading sections and deadlines: TBD.

Research practice & communication

Cite papers, figures, code and other sources; distinguish the authors' claims from your own analysis. Formal academic-integrity and AI-use policies, course forum and accessibility arrangements: TBD.

Weekly schedule

Week numbers only. Topic selections and student presenters are TBD. The candidate pool below is larger than the final course.

Week	Selected topic	Presenter
1	TBD	TBD
2	TBD	TBD
3	TBD	TBD
4	TBD	TBD
5	TBD	TBD
6	TBD	TBD
7	TBD	TBD
8	TBD	TBD
9	TBD	TBD
10	TBD	TBD
11	TBD	TBD
12	TBD	TBD
13	TBD	TBD

Suggested meeting structure

Two academic hours per selected topic. Assuming two 45-minute academic hours: 15 minutes foundations, 35 minutes mechanism and derivation, 15 minutes evidence, 25 minutes discussion. Adjust the split to the local academic-hour convention.

Choosing one focus

TTT can focus on distribution-shift adaptation and IT3, or on learning as neural memory; these are alternative focuses, not two required weeks. Drifting and IMLE can share a comparison session on one-step generation, with one method developed in depth. Selective comparisons keep either topic within one meeting.

Candidate topic pool

Choose one focused question and one main paper for each selected session. Background and recent readings supply selective preparation and comparisons. An alternative focus is a choice within the same candidate topic.

Kernels, feature learning & optimization

NTK and feature learning

When can a neural network be understood as a kernel method, and when do its features change?

Background: Lee et al., [Wide Neural Networks of Any Depth Evolve as Linear Models Under Gradient Descent](#) (2019), selected sections on Taylor linearization and squared-loss dynamics. Review Jacobians, Gram matrices and gradient flow.

Main paper: Jacot, Gabriel & Hongler, [Neural Tangent Kernel: Convergence and Generalization in Neural Networks](#) (2018).

Recent development & discussion: Haas et al., [On the Surprising Effectiveness of Large Learning Rates under Standard Width Scaling](#) (2025).

Presentation focus: Derive function-space gradient flow and its solution for squared loss, then compare a finite network with its linearization. Use the recent paper to examine why width, parameterization, learning rate and the choice of loss matter for claims about feature learning.

muP, scaling and hyperparameter transfer

How should initialization and updates scale as networks become wider or deeper?

Background: Yang & Hu, [Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks](#) (2021), selected sections on parameterization and feature learning. Review the NTK linearized regime as contrasting background; a separate NTK seminar is optional.

Main paper: Yang et al., [Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer](#) (NeurIPS 2021; arXiv 2022).

Recent development & discussion: Dey et al., [Don't Be Lazy: CompleteP Enables Compute-Efficient Deep Transformers](#) (2025).

Presentation focus: Work through activation and update scaling in a small multilayer network, and inspect a learning-rate transfer experiment. Discuss the additional constraints introduced by depth; width transfer alone does not establish depth transfer or effective nonlinear learning.

Geometry of optimization: Muon

How does the norm used to measure an update change the direction of steepest descent?

Background: Bernstein & Newhouse, [Old Optimizer, New Norm: An Anthology](#) (2024), selected sections on steepest descent, dual norms and matrix operator norms.

Main paper: Liu et al., [Muon Is Scalable for LLM Training](#) (2025).

Recent development & discussion: Kovalev, [Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization](#) (2025).

Presentation focus: Derive the spectral-norm-constrained linearized update using the SVD and explain its polar factor and Newton-Schulz approximation. Examine a scaling experiment and distinguish the geometric argument from implementation choices, weight decay and empirical comparisons with AdamW.

Sequence models & reasoning

State-space models and Mamba

How do recurrence, selectivity and structured matrices produce an efficient sequence model?

Background: Gu, Goel & Ré, [Efficiently Modeling Long Sequences with Structured State Spaces](#) (S4, 2021/22), selected sections on discretization and recurrence. Gu & Dao, [Mamba: Linear-Time Sequence Modeling with Selective State Spaces](#) (2023), selected sections on input-dependent dynamics.

Main paper: Dao & Gu, [Transformers Are SSMS: Generalized Models and Efficient Algorithms Through Structured State Space Duality](#) (Mamba-2, 2024).

Recent development & discussion: Lahoti et al., [Mamba-3: Improved Sequence Modeling Using State Space Principles](#) (2026).

Presentation focus: Unroll an SSM into a structured matrix and explain the restricted attention-SSM duality used by Mamba-2. Examine a state-tracking experiment from Mamba-3 and distinguish algorithmic complexity from measured hardware efficiency.

In-context learning as an algorithm

How can a forward pass perform learning without changing the network's parameters?

Background: von Oswald et al., [Transformers Learn In-Context by Gradient Descent](#) (2023), selected sections on the linear-attention construction. Review least squares, ridge regression and Bayesian prediction for linear models.

Main paper: Akyürek et al., [What Learning Algorithm Is In-Context Learning? Investigations with Linear Models](#) (2023).

Recent development & discussion: Wurgajt et al., [In-Context Learning Strategies Emerge Rationally](#) (2025).

Presentation focus: Work through one regression-learning construction and compare gradient-descent, ridge and Bayesian predictors on a controlled task distribution. Separate existence constructions and observed behavior in trained synthetic-task models from claims about general language models.

Test-time learning: TTT and IT3

What should a model learn at inference, and which objective makes those updates useful?

Scope for one session: Choose one route for a seminar of two academic hours: adaptation with IT3 (shown below), or learning as neural memory (alternate route). Both routes can be selected independently by different presenters if there is interest.

Background: Sun, Wang, Liu, Miller, Efros & Hardt, [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#) (ICML 2020). Shared encoders, self-supervised auxiliary tasks and the gradient-alignment condition under the paper's smooth/convex assumptions.

Main paper: Durasov, Shocher, Oner, Chechik, Efros & Fua, [IT3: Idempotent Test-Time Training](#) (ICML 2025). Learn prediction feedback and use consistency between successive predictions as the test-time objective.

Recent development & discussion: IT3 is the recent anchor. Supplementary resources: [arXiv paper](#) and [official implementation](#). Compare its idempotence-based consistency objective with the original TTT auxiliary-task design.

Presentation focus: Explain the original TTT gradient-alignment intuition, derive the IT3 consistency loss, and examine one adaptation experiment or failure case. IT3 requires its own training setup. Keep neural memory as a brief comparison; discuss when consistency is uninformative and how to stop adaptation.

Alternative focus: learning as neural memory

Background: Ba et al., [Using Fast Weights to Attend to the Recent Past](#) (2016). Fast versus slow weights and associative memory; recap the original Sun et al. TTT formulation. Original TTT: Sun et al., [Test-Time Training with Self-Supervision for Generalization under Distribution Shifts](#) (2020), selected introduction.

Main paper: Sun et al., [Learning to \(Learn at Test Time\): RNNs with Expressive Hidden States](#) (2024). A model is the recurrent hidden state, and self-supervised gradient updates are the state transition.

Recent development & discussion: Tandon et al., [End-to-End Test-Time Training for Long Context](#) (2025). Next-token updates compress context into weights, with a meta-learned initialization. Brief contrast: [IT3](#) (2025), adaptation under distribution shift.

Presentation focus: Work through a TTT-Linear update and its recurrent computation. Compare layer-local reconstruction with end-to-end next-token adaptation; distinguish sequence memory from adapting to a shifted test distribution. This is a complete alternative seminar of two academic hours, rather than a required continuation of the adaptation route.

RL for reasoning with verifiable rewards

Does reward-based training discover successful reasoning behavior or change which existing behavior is sampled?

Background: [Spinning Up: Part 3-Introduction to Policy Optimization](#), selected sections on the log-derivative policy gradient and baselines. Reserve 25-30 minutes to introduce policies, sampled completions, rewards and group-relative advantages; prior RL coursework is not assumed.

Main paper: DeepSeek-AI et al., [DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning](#) (2025), focusing on R1-Zero and the learning objective.

Recent development & discussion: Yue et al., [Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?](#) (2025). Counterpoint: Liu et al., [ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models](#) (2025).

Presentation focus: Derive the policy-gradient identity before introducing group-relative advantages, clipping and KL regularization. Compare pass@k evidence under the papers' different training and evaluation settings; finite sampling cannot establish that a base model assigns zero probability to a solution.

Recurrent depth and latent reasoning

Can a model gain reasoning ability by repeatedly applying the same layers?

Background: Dehghani et al., [Universal Transformers](#) (ICLR 2019; 2018 preprint). Shared layers, recurrence over depth and adaptive computation.

Main paper: Geiping et al., [Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach](#) (NeurIPS 2025).

Recent development & discussion: Jolicoeur-Martineau, [Less is More: Recursive Reasoning with Tiny Networks](#) (2025). A short comparison with recursive refinement on puzzles.

Presentation focus: Trace one recurrent block, explain how it is trained across iteration counts, and assess performance against inference compute. Use tiny recursive models as a comparison. Distinguish iteration count, parameter count and demonstrated reasoning performance.

Adaptation, representations & world models

Hypernetworks and generated adaptation

Can a network generate the parameters another network needs for a new task or context?

Background: Ha, Dai & Le, [HyperNetworks](#) (2016/17). Hu et al., [LoRA: Low-Rank Adaptation of Large Language Models](#) (2021/22), selected sections on low-rank parameter updates.

Main paper: Charakorn et al., [Text-to-LoRA: Instant Transformer Adaption](#) (2025).

Recent development & discussion: Charakorn et al., [Doc-to-LoRA: Learning to Instantly Internalize Contexts](#) (2026).

Presentation focus: Explain the conditioning signal, generated low-rank update and training objective, then examine a held-out-task or document experiment. Compare the cost and information limits of generated adaptation with fine-tuning and keeping the original context available.

JEPA and predictive world models

When does prediction in representation space support understanding, dynamics and planning?

Background: Assran et al., [Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture](#) (I-JEPA, 2023), selected sections on masking, target representations and collapse prevention.

Main paper: Assran et al., [V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning](#) (2025).

Recent development & discussion: Maes et al., [LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels](#) (2026).

Presentation focus: Trace the representation-prediction objective and action-conditioned planning loop, then analyze a planning result and its data requirements. Contrast target-network stabilization with regularizing the latent distribution; representation quality and accurate action-conditioned dynamics are separate questions.

Representation alignment and the Platonic hypothesis

Do different models learn the same representations, and how would we know?

Background: Kornblith et al., [Similarity of Neural Network Representations Revisited](#) (ICML 2019). Centered kernel alignment and the invariances of similarity measures.

Main paper: Huh et al., [Position: The Platonic Representation Hypothesis](#) (ICML 2024).

Recent development & discussion: Gröger, Wen & Brbic, [Revisiting the Platonic Representation Hypothesis: An Aristotelian View](#) (ICML 2026). Calibration for similarity scores and the distinction between global similarity and local neighborhood agreement.

Presentation focus: Compute one similarity measure on a small example, examine a cross-modal alignment result, and test its interpretation against the recent calibration critique. Treat convergence to a shared reality as a hypothesis whose evidence depends on the measurement.

Editing & interpreting networks

Model merging and task arithmetic

Why can adding parameter updates combine capabilities, and when does task interference prevent it?

Background: Ilharco et al., [Editing Models with Task Arithmetic](#) (2023). Review parameter differences relative to a shared pretrained checkpoint and function-space linearization from NTK background.

Main paper: Ortiz-Jimenez, Favero & Frossard, [Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models](#) (2023).

Recent development & discussion: Hitit, Gırrbach & Akata, [A Systematic Study of In-the-Wild Model Merging for Large Language Models](#) (2025 preprint; TMLR 2026).

Presentation focus: Derive how adding task vectors affects a linearized model and examine the proposed weight-disentanglement mechanism. Compare controlled task combinations with heterogeneous checkpoints, including conflicting objectives and gains relative to the best individual model.

Mechanistic interpretability: features and superposition

What is a feature, and can sparse autoencoders recover the units a network computes with?

Background: Elhage et al., [Toy Models of Superposition](#) (2022), selected sections on sparse features, interference and the two-feature toy model.

Main paper: Templeton et al., [Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet](#) (2024).

Recent development & discussion: Leask et al., [Sparse Autoencoders Do Not Find Canonical Units of Analysis](#) (2025).

Presentation focus: Explain the sparse-autoencoder objective and reproduce or analyze a small superposition experiment. Examine feature interventions alongside the stitching and meta-SAE evidence, asking whether interpretability, completeness and atomicity are being measured separately.

Mechanistic interpretability: circuits and causal tests

How can we trace a model's computation and test whether the proposed explanation is causal?

Background: Elhage et al., [A Mathematical Framework for Transformer Circuits](#) (2021), selected sections on the residual stream and attention's QK/OV decomposition. Review sparse representations from the features background; a separate features seminar is optional.

Main paper: Ameisen et al., [Circuit Tracing: Revealing Computational Graphs in Language Models](#) (2025).

Recent development & discussion: Kamath et al., [Tracing Attention Computation Through Feature Interactions](#) (2025).

Presentation focus: Explain how a replacement model produces an attribution graph and work through one circuit case study with an intervention. Identify approximation errors and omitted computations, then examine what the attention extension adds to the explanation.

Generative models

One-step generative models: Drifting and IMLE

How can matching real and generated samples train a generator that produces a sample in one pass?

Scope for one session: One comparative seminar of two academic hours. Use Drifting for the central derivation and IMLE / ROMS-IMLE for comparison, or choose the IMLE-first route below. Separate deeper seminars are available if both focuses are selected.

Background: Yujia Li, Swersky & Zemel, [Generative Moment Matching Networks](#) (ICML 2015), selected kernel/MMD material. Ke Li & Jitendra Malik, [Implicit Maximum Likelihood Estimation](#) (2018), the data-to-generated nearest-neighbor assignment and regression algorithm.

Main paper: Deng, He Li, Tianhong Li, Du & He, [Generative Modeling via Drifting](#) (2026). Estimate an attraction-repulsion field from real and generated samples and fit frozen drifted targets.

Recent development & discussion: Chirag Vashist & Ke Li, [ROMS-IMLE: A Minimalist Approach to Competitive Single-Step Generative Modelling](#) (2026). Compare nearest-neighbor assignment, robust losses and intermediate-resolution supervision with the drifting update. Optional theory: Esteban-Casadevall et al., [Kernel-Gradient Drifting Models](#) (2026), including Max Welling.

Presentation focus: Introduce sample-based distribution matching, derive the drifting field and frozen-target regression, and study one toy experiment. Compare IMLE's data-to-generated matching and ROMS-IMLE's practical changes briefly. Choose one main mechanism to derive; keep zero-drift and maximum-likelihood claims tied to their assumptions.

Alternative focus: IMLE and mode coverage

Background: Ke Li & Jitendra Malik, [Implicit Maximum Likelihood Estimation](#) (2018). Implicit distributions, nearest-neighbor assignment, mode coverage and the likelihood connection under stated assumptions.

Main paper: Chirag Vashist & Ke Li, [ROMS-IMLE: A Minimalist Approach to Competitive Single-Step Generative Modelling](#) (2026). Revisit IMLE with intermediate-resolution supervision and robust losses.

Recent development & discussion: ROMS-IMLE is the recent anchor. The authors' [method page](#) provides additional explanation and experiments. Comparison: Deng et al., [Generative Modeling via Drifting](#) (2026), selected algorithm and toy example.

Presentation focus: Explain the direction of nearest-neighbor matching, latent assignment and alternating regression, then examine sampling cost and the ROMS-IMLE ablations. Keep the maximum-likelihood connection tied to its assumptions. Use Drifting as a brief comparison within the same two-academic-hour seminar.

Diffusion language models

How can denoising generate discrete text, and what changes when generation need not proceed left to right?

Background: Austin et al., [Structured Denoising Diffusion Models in Discrete State-Spaces](#) (2021), selected sections on transition matrices and absorbing-mask corruption.

Main paper: Sahoo et al., [Simple and Effective Masked Diffusion Language Models](#) (2024).

Recent development & discussion: Nie et al., [Large Language Diffusion Models](#) (LLaDA, 2025).

Presentation focus: Derive the masked-denoising objective and trace one sampling schedule, then compare refinement, parallelism and likelihood evaluation with autoregressive generation. The topic requires discrete corruption and language-specific evaluation beyond a continuous diffusion refresher.

Symmetry & alternative dynamics

Symmetry, equivariance and canonicalization

Can we enforce symmetry by putting each input into a canonical orientation?

Scope for one session: Advanced mathematical candidate. A concrete rotation or permutation example is enough to motivate the main result within two academic hours.

Background: Cohen & Welling, [Group Equivariant Convolutional Networks](#) (ICML 2016). Group actions, equivariance and weight sharing.

Main paper: Dym, Lawrence & Siegel, [Equivariant Frames and the Impossibility of Continuous Canonicalization](#) (ICML 2024).

Recent development & discussion: Shumaylov et al., [Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups](#) (ICLR 2025; 2024 preprint). Selected introduction and one application.

Presentation focus: Start with a group-action example, show why a canonical choice can become discontinuous under the stated assumptions, and explain how weighted frames address continuity. Leave general topology proofs and non-compact Lie-group machinery for optional reading.

Oscillatory neurons and synchronization

Can neural representations bind information through synchronization?

Scope for one session: Exploratory candidate connecting neural computation with dynamical systems, including work by Max Welling.

Background: Strogatz, [From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators](#) (2000). Sections 2 and 3.1-3.3 on coupled phases and synchronization (author-hosted PDF).

Main paper: Miyato, Löwe, Geiger & Welling, [Artificial Kuramoto Oscillatory Neurons](#) (ICLR 2025; 2024 preprint).

Recent development & discussion: Iqbal et al., [Spontaneous symmetry breaking and Goldstone modes for deep information propagation](#) (2026 preprint). A related perspective on symmetry and persistent information propagation, rather than a direct analysis of AKOrN.

Presentation focus: Derive the two-oscillator intuition, trace the generalized neural update, and study one object-discovery or reasoning experiment. Use the Goldstone paper for a brief connection to depth and memory; its physics derivations are optional.

Course website: [Deep Deep Learning](#)